

Cost-Efficient LLM Bakeoff for Agentic Coding

Selecting a lower-cost model on AWS Bedrock to replace Claude Sonnet 5

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Executive Summary

- **Model comparison:** Gemma 4 31B and Nemotron 3 Super 120B are cheapest (~95-96% savings); Kimi K2.5 leads on vendor-reported benchmarks but saves less (~80-82%).
- **Model comparison:** Vendor vs. independent benchmarks disagree sharply (Kimi's SWE-bench lead reverses under independent testing), so a blind internal eval on real repos decides.
- **LiteLLM vs. OpenRouter:** Route via a self-hosted LiteLLM proxy calling Bedrock directly - OpenRouter's Bedrock path still transits their servers plus a ~5% fee.
- **Timeline:** ~2-2.5 week bakeoff (build eval, run, score, recommend), then a 10-20% canary rollout with Sonnet 5 fallback.

Recommendation

Our agentic coding workloads (project/feature-scale, multi-file work) currently run primarily on Claude Sonnet 5. Advance **Nemotron 3 Super 120B**, **Gemma 4 31B**, and **Kimi K2.5** to a ~2 week bakeoff on real Pindrop tasks. Nemotron is the cheapest of the three after Gemma and has the largest output ceiling, though its benchmark coverage leans more heavily on vendor-reported numbers than the other two; Kimi leads on the one benchmark it publishes itself (SWE-bench Verified, unverified) but trails on the one independently-run benchmark that covers it; Gemma leads on cost, license, and the independent benchmark data that exists, though it's the newest of the three on Bedrock. All three cut spend 80%+ vs. Sonnet 5, so the decision should turn on measured task quality on our own repos, not price or benchmark scores.

Candidates at a Glance

Model	SWE-bench Verified	Terminal-Bench Hard (agentic CLI, independent)	Bedrock price (in/out, per 1M)	Context / Max output
Kimi K2.5	76.8% (vendor)	34.8%	~\$0.60 / \$2.50–3.00	256K / 16K
Nemotron 3 Super 120B	60.5% (vendor)	25.8% (vendor)\$	\$0.15 / \$0.65	256K / 32K
Gemma 4 31B	Not published	36.4%	\$0.14 / \$0.40	256K / not disclosed
Claude Sonnet 5 (reference)	85.2% (vendor)†	Not published	\$3.00 / \$15.00 (standard)	1M / 128K

SWE-bench Verified scores exist for Kimi, Nemotron, and Sonnet 5 (published by Moonshot, NVIDIA, and Anthropic respectively; NVIDIA is the only one of the three to disclose multiple harness variants, 53.7-60.5% depending on harness, rather than one cherry-picked number) - Gemma has none published anywhere we could find. Treat all published scores with a discount: OpenAI stopped reporting this benchmark for frontier evals in Feb 2026 after an audit of 138 tasks found ~59% had material test/problem-design flaws (the single largest category, overly narrow test cases, was ~35% on its own) and every frontier model tested had memorized the human-written fix - it's the most contamination-prone number in this table, included because it's the only benchmark with any real coverage, not because it's trustworthy.

Artificial Analysis Index Scores

Independent, composite scores from Artificial Analysis - a different lens than the task-specific benchmarks above, so kept in its own table rather than mixed in:

Model	AA Intelligence Index	AA Coding Index	Output speed (tok/s)
Kimi K2.5	38	46.8	49.9
Nemotron 3 Super 120B	25	37.7	148.9
Gemma 4 31B	29.4	43.4	35.3
Claude Sonnet 5 (reference)	53	71.5	74.3

Cost Comparison

At ~665M input + 200M output tokens/month (volume rescaled so the Sonnet 5 baseline lands at \$5,000/month):

Model	Price (in/out, per 1M)	Est. monthly cost	Savings vs. Sonnet 5
Gemma 4 31B	\$0.14 / \$0.40	~\$175	~96%
Nemotron 3 Super 120B	\$0.15 / \$0.65	~\$230	~95%
Kimi K2.5	~\$0.60 / \$2.50–3.00	~\$900–1,000	~80–82%
Claude Sonnet 5	\$3.00 / \$15.00	~\$5,000	baseline

Artificial Analysis found Sonnet 5's cost *per completed task* runs ~15% higher than Opus 4.8's despite its lower per-token price, because it uses more tokens per task - a reminder that cost per completed workflow, not per token, should decide this bakeoff.

Model Trade-offs

- **Kimi K2.5** - leads on the vendor-reported SWE-bench Verified figure (76.8%) and has the highest AA Intelligence Index and AA Coding Index of the three candidates (38 and 46.8), but the one time an independent evaluator actually tested it (Terminal-Bench Hard), it scored 34.8% - below Gemma's 36.4% and well below its own vendor-claimed 50.8% on a different Terminal-Bench track. Treat its self-reported lead as a hypothesis, not a fact - this is direct evidence, not just a caveat, that vendor numbers and independent numbers can disagree by double digits. 16K max output, the smallest disclosed of the three (Gemma's isn't disclosed); watch for truncation on large diffs. Modified MIT license: above 100M MAU or \$20M/month revenue, "Kimi K2.5" must be displayed in-product - not a blocker at Pindrop's scale.
- **Nemotron 3 Super 120B** - cheapest after Gemma (\$0.15/\$0.65), and the largest output ceiling of the three (32K). Weakest vendor-reported SWE-bench Verified of the three (60.5%), though NVIDIA is the only vendor here to disclose multiple harness variants rather than one cherry-picked number. AA Intelligence Index (25) and AA Coding Index (37.7) are both genuinely published - though its Terminal-Bench Hard figure (25.8%) is vendor-reported, not independently run like Gemma's and Kimi's. NVIDIA Nemotron Open Model License: no revenue or MAU threshold turned up in what we reviewed, but confirm full terms with legal like the others.
- **Gemma 4 31B** - cheapest, cleanest license (Apache 2.0, no thresholds), strong generalist with native function calling, and now the leader among the three candidates on both independent benchmarks that exist for it: AA Coding Index (43.4) and Terminal-Bench Hard (36.4%). No proven repo-issue-resolution track record - no SWE-bench Verified score exists for it. Newest on Bedrock (June 2026), so least production history of the three; served via the bedrock-mantle endpoint rather than the standard bedrock-runtime endpoint, which is why it may not appear under the console's Serverless model-catalog filter.

Model Access Layer: LiteLLM vs. OpenRouter

LiteLLM as proxy. LiteLLM is an open-source gateway we can self-host that gives us one unified way to call and switch between all the candidate models, plus built-in cost tracking. Since it runs inside our

own AWS account and talks to Bedrock directly, none of our data passes through a third-party vendor.

Why not OpenRouter. OpenRouter now offers a “Bring Your Own Key” mode that can reach a customer’s own Bedrock account, but under BYOK requests are still proxied through OpenRouter’s own servers before reaching Bedrock, and OpenRouter charges roughly 5% on top of Bedrock’s normal cost. For a shop standardized on Bedrock for pricing and data-residency reasons, routing traffic through a third-party vendor’s servers - even briefly, even under BYOK - cuts against the reason we’re on Bedrock in the first place, so OpenRouter isn’t being evaluated as the routing layer.

Eval Approach

No public benchmark has full, contamination-clean coverage of all three candidates, so a two-tier internal eval set on real Pindrop repos is the primary decision input:

- **Tier A - bug fix** (~20–30 tasks): 1–2 files, clear pass/fail.
- **Tier B - feature scale** (~20–30 tasks): multi-file implementations/refactors, graded on cross-file consistency and end-to-end correctness. This is where the recommendation should actually be won or lost.

Grade on: task completion, tool-execution accuracy, cross-file consistency, tests passing, reviewer pass-through rate, latency, and **cost per completed workflow** (not per token). N≥3 samples per task, graded blind.

Success criteria: task completion ≥90% of Sonnet 5 (weighted to Tier B); terminal-command success ≥95%; cross-file changes correct ≥90%; cost ≤40% of Sonnet 5 per completed workflow.

Timeline (~2.5–3 Weeks)

- **Phase 0 (~1 day):** Enable Bedrock model access for all three (auto-subscribe, allow ~15 min propagation), confirm region/bedrock-manage endpoint per model, check Service Quotas (a quota increase can take ~1 business day), and re-check SWE-rebench and AA’s Terminal-Bench Hard leaderboard for any newly-published independent scores on Nemotron or Sonnet 5.
- **Phase 1 (~1-1.5 weeks):** Build the two-tier eval set; run all three models N≥3 samples/task, graded blind.
- **Phase 2 (~2–3 days):** Score against success criteria; compute cost per completed workflow; compare to the Sonnet 5 baseline.
- **Phase 3 (~2 days):** Recommend one model; plan a 10–20% canary rollout with Sonnet 5 fallback.

Decisions Needed

- Approve the candidate lineup (Nemotron 3 Super 120B, Gemma 4, Kimi K2.5) and the ~2.5–3 week timeline.

- Confirm Bedrock access, IAM permissions, and any needed Service Quota increases for the three models.
- Approve the 40–60 task two-tier eval set and the success thresholds above.
- Confirm Sonnet 5 (not Opus 4.8) is the correct baseline, and which pricing tier - standard (\$3.00/\$15.00, used throughout this report)
- Legal sign-off on Kimi's modified-MIT license and Nemotron's Nemotron Open Model License terms before production rollout.

Sources

Nemotron 3 Super 120B - SWE-bench Verified (60.5%, OpenHands harness; 53.7–60.5% range across harnesses) and vendor Terminal-Bench Hard figure (25.8%): [NVIDIA Nemotron 3 Super 120B model card](#) - Conflicting Terminal-Bench Hard figure (29%): [Artificial Analysis — Nemotron 3 Super article](#) - Bedrock pricing (\$0.15/\$0.65), context/max output (256K/32K), model ID: [AWS Bedrock model card](#) · [AWS Bedrock pricing](#) - AA Intelligence Index (25) and output speed (148.9 tok/s): [Artificial Analysis model page](#) - AA Coding Index (37.7): [BenchLM.ai AA Coding Index leaderboard](#) - License terms: [NVIDIA Nemotron Open Model License](#) - Bedrock GA date (March 2026): [AWS What's New](#)

Kimi K2.5 - SWE-bench Verified (76.8%) and vendor Terminal-Bench 2.0 figure (50.8%): [Kimi K2.5 GitHub](#) · [Hugging Face model card](#) - Independent Terminal-Bench Hard figure (34.8%): [Artificial Analysis Terminal-Bench Hard leaderboard](#) - Bedrock pricing, context/max output, model ID, Bedrock launch date (Jan 2026): [AWS Bedrock model card](#) - AA Intelligence Index (38) and default output speed (49.9 tok/s): [Artificial Analysis model page](#) - AWS Bedrock-specific output speed (118 tok/s): [Artificial Analysis providers page](#) - License terms (100M MAU / \$20M revenue threshold): [Kimi K2.5 LICENSE](#)

Gemma 4 31B - Terminal-Bench Hard (36.4%): [Artificial Analysis — Gemma 4 article](#) - Bedrock pricing, context, model ID, native function calling: [AWS Bedrock model card](#) · [AWS Bedrock pricing](#) - AA Intelligence Index (29.4) and output speed (35.3 tok/s): [Artificial Analysis model page](#) - License (Apache 2.0): [Google's Gemma 4 announcement](#) - Bedrock GA date (June 2026): [AWS What's New](#)

Claude Sonnet 5 - SWE-bench Verified (85.2%) and the Pro-vs-Verified mislabeling risk (Pro = 63.2%): [Vellum — Sonnet 5 benchmarks explained](#) · [Apidog — Sonnet 5 benchmarks](#) - Independent Terminal-Bench v2.1 figure (74.53%): [vals.ai Terminal-Bench 2.1 leaderboard](#) - Vendor Terminal-Bench figure (80.4%): [Vellum — Sonnet 5 benchmarks explained](#) - Bedrock standard + introductory pricing, context/max output, model ID: [AWS Bedrock model card](#) · [AWS launch blog](#) - AA Intelligence Index (53) and output speed (74.3 tok/s): [Artificial Analysis model page](#) - AA Coding Index (71.5): [BenchLM.ai Sonnet 5 page](#) - Cost-per-completed-task finding (~15% above Opus 4.8): [Artificial Analysis — Sonnet 5 agentic cost article](#)

Model Access Layer - LiteLLM Bedrock provider support, self-hosted proxy/gateway architecture: [LiteLLM — Bedrock provider docs](#) · [LiteLLM — Proxy Server docs](#) · [LiteLLM — Proxy reliability/fallback docs](#) · [BerriAI/litellm GitHub](#) - OpenRouter Bring-Your-Own-Key support for Amazon Bedrock: [OpenRouter — BYOK use case docs](#) · [OpenRouter — BYOK auth guide](#) · [OpenRouter — Amazon Bedrock provider page](#) - OpenRouter BYOK data path (traffic proxied through OpenRouter's servers before reaching

the customer's Bedrock account): stormacq.com — Xcode + OpenRouter + Bedrock walkthrough - OpenRouter BYOK ~5% fee on top of underlying provider cost: [OpenRouter Help Center](#) — BYOK billing

Methodology / cross-cutting - OpenAI's SWE-bench Verified deprecation and audit findings (~59% of tasks flawed, contamination evidence): [OpenAI — Why we no longer evaluate SWE-bench Verified](#) - Terminal-Bench 2.0 / 2.1 / Hard as distinct, non-comparable tracks: tbench.ai — Terminal-Bench 2.1 announcement · [tbench.ai benchmarks](https://tbench.ai/benchmarks) - AA Intelligence Index composition (9 evaluations): [Artificial Analysis methodology](#) - AA Coding Index composition (Terminal-Bench v2.1 + SciCode): [Artificial Analysis coding capabilities page](#)